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St. Francis E.M.B High School, Machilipatnam Andhra Pradesh



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Title/Name of the project: 'PRAGATHI' A survey to analyze relation between academics, co-curricular activities and student well-being

Focal theme: "Understanding Ecosystem for health and wellbeing"

Sub theme: "2. Fostering Health, Nutrition and well-being"

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*“The highest education is that which does not merely give us
Information but makes our life in Harmony with all
Existence”*

GURUDEV
-RABINDRANATH TAGORE-



INDEX

1	Abstract.....	4		
2	Key words.....	4		
3	Introduction.....	5		
4	Hypothesis	6		
5	Rationale of the study	6		
6	Objectives	6		
7	Methods & methodology	7		
7.1	History	8		
7.2	Parameters under study.....	10		
7.2.1	Stress (Mental Health)	10		
7.2.2	INTENSITY OF STRESS	11		
7.2.3	REASONS	11		
7.2.4	impact of stress on student's life	11		
7.3	Physical Health	12		
	7.4 ACEDAMIC PERFORMANCE	12		
	7.5 Co-curricular activities & others	12		
8	SAMPLE SIZE	13		
9	Data analysis	14		
	9.1 DEVLOPING MHI.....	14		
	9.2 DEVLOPING PHI.....	15		
	9.3 DEVLOPING CAI.....	17		
	9.4 DEVLOPING SWI.....	17		
10	Results	18		
11	Discussion of results & Future scope.....	21		
12	Limitations.....	21		
13	CONCLUSION	21		
14	Acknowledgement.....	22		
15	References.....	23		

1 Abstract

well-being(noun)/ˌweɪlˈbiːŋ/, *a state of being comfortable, healthy & happy*[1]. Well-being of students has become a matter for concern for both parents and teachers alike. Within the context of current competitive world and various emerging fields, our school St. Francis E.M.B high school, Machilipatnam deals with large number of students from diverse backgrounds and a blend of nearly all cultures. As we firmly believe that “*every child is special*” it is impossible to even estimate the well-being of student just by looking upon the marks. Under this situation we require a method to determine the wellbeing of student. In this approach a survey is conducted with students of classes 9th and 10th. Primarily questions which determine stress experienced by student, participation in cultural activities & Physical health of student were asked. Based on the observations made we have developed a grading system to include in progress reports which not only has the percentage of marks secured but also Mental Health Index (MHI), Physical Health Index (PHI) & Cultural Actives Index (CAI) which enables the parents and teachers to evaluate the performance of child more accurately. with data obtained many observations were noted of which include optimum time for a child to give for his hobby or sports, intensity of mental stress experienced by students, Reasons for the stress, trends in mental health, academic performance, hobby time and academic performance

with the help of data collected finally the results are analyzed and conclusions are drawn

2 Key words

Progress reports, Mental stress, Student well-being, School ecosystem

3 Introduction

In the era of rat race and cut throat competition the mental health of student has become a matter of concern. Though we firmly believe “every child is special” i.e., everyone has their own passion and goals but same syllabus! This increases stress and mental pressure on students. To get to know more about this issue and hopefully a solution we introduce you to our project

‘PRAGATI’

A survey to analyze relation between curricular, co-curricular activities and student wellbeing.

The survey is our only way to explore the deep roots of stress, so we conducted a survey.

- 1) Surveys of this kind had been done but on university students and Graduates. But not on school level. As school students are much sensitive than University students and graduates it is difficult to cope up for them.[5]
- 2) surveys done till now were in western countries and not suitable for Indian school ecosystem
- 3) A survey is conducted where sample size is very diverse as students from class 9th & 10th participated in it with large sample size of nearly with 250 students.
- 4) We all know that issues such as stress and mental health can't be understood by survey of students alone, we have also included the information about academic progress of students from the teacher's desk through their progress cards.

Main aim of the survey is to revolutionize progress reports to ‘Energized progress reports’ which include Mental Stress index (MSI), Disciplinary Index (DI) along with academic performance. To keep a check on overall development of student, Student Wellness index (SWI) is published which is a blend of all data collected

I hope that this attempt changes the perspective of Parents, students & Teachers regarding progress report and academics.

4 Hypothesis

- Students who have hobbies are less likely to experience academic stress than students who do not have
- Students who engage in hobbies have higher levels of mental well being
- The results of the survey enable parents understand their children's overall development
- This survey also enables the parents to know the children's interest in particular field

5 Rationale of the study

School as a whole is big ecosystem and all students, teachers & parents are part of it. But one of the major problems faced by all of the groups was the discontentment of Students. This project aims to analyze student wellbeing and relate this to academic performance of students. As most of the schools tend to ignore this issue this projects sheds light on it enables parents and teachers understand students much better.

6 Objectives

Main objectives of this project are:

- To learn about stress levels in students.
- To Revolutionise progress reports.
- Analyse trends between curricular data(marks) and co-curricular activities.
- To change the perspective of Parents, Students alike regarding progress report.
- To develop an evaluation tool for a better self-evaluation of students.

7 Methods & methodology

As research plays a major role in any project making it more accurate. So, more time is allotted to research in this project. The research is carried out on following topics

- History of project
- Parameters under study
- Sample size
- Data analysis

This project makes free use of abbreviations as stated below Abbreviation table

Sc no.	Long form abbreviation	Abbreviation
1	Mental Health Index	MHI
2	Physical Health Index	PHI
3	Cultural Activities Index	CAI
4	Student Wellness Index	SWI
4	Body Mass index	BMI

TABLE 1

7.1 History

To get a clear idea about the

- Sample size
- Factors affecting stress
- Hobbies
- Statistics to evaluate

Intense research is demanded by the project. The trends, methods of collecting data, data analysis can be understood by looking to surveys of similar kind done in the past, those were found in review paper as the listed below [5].

No	Researchers	Source of academic stress	Research Participant
1.	Lee, Kang, & Yum (2005)	a) score and competition, b) career and future success, c) various demands and deadlines [15]	Undergraduate students in Korea
2.	Yan & Berliner (2009)	a) cultural and educational differences, b) limited language skills, c) ineffective interactions with American students [16]	International undergraduate students from China studying in the USA
3.	Agolla & Ongori (2009)	a) academic overload, b) fear of failure) inadequate facilities (computers, books, lectures) [12]	Undergraduate students in Botswana
4.	Kausar (2010)	Academic workload [14]	Undergraduate students in Punjab, Pakistan
5.	Tan & Yates (2011)	Parent, teacher, and self-academic expectation [13]	Senior high school students in Singapore
6.	Nandamuri & Gowthami (2012)	Daily assignments in class, placement activities, assessment, and group assignments [17]	Students of undergraduate management department in Andhra Pradesh, India
7.	May & Casazza, 2012)	Difficult courses (pharmacy and engineering) [18]	Undergraduate students in third, fourth, and fifth years.
8.	Weerasinghe, Batagoda, Chandrika, & Siriwardana, 2012)	Academic workload [19]	Undergraduate students at University of Colombo Sri Lanka
9.	Pandya, Deshpande, & Karani (2012)	Inadequate time for recreation and lecturers' teaching strategies [20]	Undergraduate Students of MBA in Gujarat, India
10.	Calaguas (2012)	Demands from parents, teachers, and students (a person's self) [21]	Undergraduate Students in Filipina
11.	Ford, Olotu, Thach, Roberts, & Davis (2014)	Assignment load [22]	Doctoral students of Pharmacy in the USA
12.	(Nakalema & Ssenyonga, 2014)	Financial problems, academic burdens or excessive assignments with the limited deadline, and high social demands [23]	Undergraduate students in Uganda
13.	Rakhmawati, Farida, & Nurhalimah (2014)	Academic expenses (assignments, examinations, and lecture activities) [24]	Undergraduate nursing students in Jakarta

TABLE 2

Table 2 shows that most studies involve undergraduate students (85%), two studies involve senior high students, and only 1 study involves junior high school students. The limited studies on the phenomena and sources of academic stress, especially for the junior high level, the early adolescent development, might have some impacts on the lack of reference for academic stress in this age group. It makes the treatments for the students ineffective. In fact, early adolescence is a transition period with a rapid process of change in cognitive, social, and emotional aspects accompanied by various complex demands and pressures in the case of social and academic areas (Santrock, 2002). This period is when academic stress most likely occurs. In early adolescence, anxiety begins to appear, especially related to their future, and it can be manifested in the form of fear of failure at school because school achievement is considered as a determining factor for an individual's career (Wrzesniewski & Chylinska, 2007). It should be noted that various studies indicate that most psychological problems in adulthood are derived from childhood and adolescence periods (Kim-Cohen et al., 2003). Therefore, studies involving teen is critical. Concerning this unclear academic stressor condition, this study examines academic stress, its intensity, and how junior high school students deal with it. Moreover, this study investigates the things perceived by junior high school students as sources of academic stress. The results of this study are expected to help teachers and schools appropriately overcome problems caused by student academic stress.

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7.2 Parameters under study

After examining the surveys done in the past. The task of preparing the survey sheet lies ahead to prepare this we have divided questions into 4 categories

- 1) Stress
- 2) Physical health
- 3) Academic performance
- 4) Co-curricular activities & others

The blue print of the questionnaire is as follows:

Sc.NO	Aspect
1	Intensity of stress, Reasons/sources of stress, effect on daily life
2	Students' participation in school activities, hobbies
3	Height and weight, frequency of visiting doctor
4	Percentage achieved by the student in exams conducted in school

TABLE 3

7.2.1 Stress (Mental Health)

To understand mental health of student we need to understand 3 aspects

- Intensity of stress
- Reasons of stress
- Impact of stress on students' life

7.2.2 INTENSITY OF STRESS

Intensity of stress is how much stress student is experiencing in his/her daily life. As everyone has different opinions, single question cannot determine this so 3 questions are asked and average of these will be taken as Mental Health Index (MI) they are

- 1) On scale of 1 to 10 how much stress do you feel regarding academics
- 2) In the last month how often have you been able to control irritations.
- 3) I have difficulty in concentrating class because of stress

7.2.3 REASONS

Reasons of the stress is asked through a simple question. This is asked because by knowing this teachers, parents & student can do further evaluation

What do you think is main reason for stress you have experienced

- 1) Work Load 2) Exams 3) Parents expectations 4) No stress 5) Others: _____

7.2.4 IMPACT OF STRESS ON STUDENT'S LIFE

It is studied through questions: -

Are you comfortable discussing about Stress with your Parents or teachers

I have difficulty concentrating in class because of stress

in the last month how often have you been able to control irritations

7.3 Physical Health

This is measured by obtaining height and weight of student and calculating Body mass index. this also depends on how frequently a student is getting sick or visiting a doctor. This data is obtained through questions

- 1) Height
- 2) Weight
- 3) How frequently you are visiting a doctor or get sick

7.4 ACEDAMIC PERFORMANCE

- 1) The academic performance is estimated from the percentage of the student achieved in his/her school examinations. This is obtained by question-What is your percentage in last examination?

7.5 Co-curricular activities & others

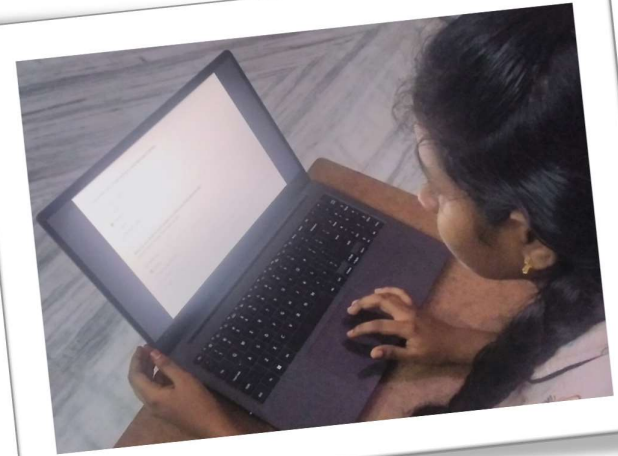
Others questions related to student activities are also asked like: -

- 1) On scale of 1 to 5, according to you how disciplined are you in school premises.
- 2) Are you having a hobby or a favorite sport?
- 3) If you have hobby or sport, how much time would you give for it on a week's span? (In hours).
- 4) How often would you participate in games or cultural events organized in school?

8 SAMPLE SIZE

Sample size of this project includes:

- Class 10th students are divided into 3 sections (A, B, C) estimating around 100 students
- Class 9th students are divided into 5 sections (A1, A2, A3, A4, A5) estimating around 150 students
- Expected sample size is 200-250 out of which we have received 230 responses



8.Datacollection

Due to large sample size data collection is done through google forms and for the ease of calculations & analysis is done using Microsoft Excel. The group leader along with the guide teacher explained the idea of the project to all students and encouraged them to participate actively. Students has given their valuable insights and feedback on topic.



9 Data analysis

The data is analyzed in excel by converting data input into numeric format and then analyzed to develop formulas



Input: is taken from the forms the students have submitted

9.1 DEVELOPING MHI

1. Intensity of stress is already taken in numerical format on lentil scales so there is no need to convert that.
2. The question How often are you be able to control is taken in words so it is converted as in table below

Statement	value
never	2.50
almost never	2.25
sometimes	2
fairly often	1.75
very often	1.25

TABLE 4

3. The question “I have difficulty concentrating in class because of stress.” Is converted as stated below

Statement	value
Strongly disagree	0.75
Disagree	0.25
Neutral	1
Agree	1.25
Strongly agree	1.75

TABLE 5



9.2 DEVELOPING PHI

1. Body mass index is calculated by formula

$$BMI = \frac{wt}{H^2} \quad \text{wt.}=\text{weight of student , H= height of student}$$

Based on this we have got physical condition of child and converted that into our scale as below. Body Mass Index (BMI) is a measure of body fat based on your weight and height. It is calculated by dividing your weight in kilograms by the square of your height in meters. The BMI categories and their corresponding values are:

BMI Range	BMI Category	SCALE VALUES
< 18.5	Underweight	0.5
18.5 – 24.9	Normal weight	1
25 – 29.9	Overweight	0.25
>= 30	Obesity	0.75

TABLE 6

2. The frequency of visiting doctor is evaluated as shown below

Statement	value
once a month	0
once two months	2
once three months	4
once four months or more	6

TABLE 7

BODY
MASS
INDEX



FREQUENCY
OF GETTING
SICK



PHYSICAL
HEALTH
INDEX

9.3 DEVELOPING CAI

CAI is calculated based on students' participation in co-curricular activities and time allotted for hobby

1. As participation in co-curricular activities is already taken in lentil scale it eliminates the need of converting it
2. Hobby time is also in numerical format (in hours).

CAI is simply calculated by adding both parameters



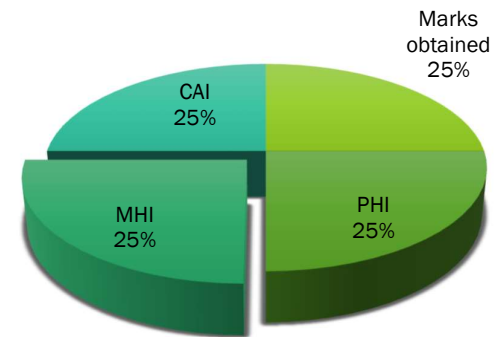
9.4 DEVELOPING SWI

The final index which determines the overall wellbeing of student is SWI it is shown in percentage higher is the SWI higher is wellbeing of student the weightage of all parameters in SWI is as follows

SWI is developed by adding all quantities and obtaining percentage



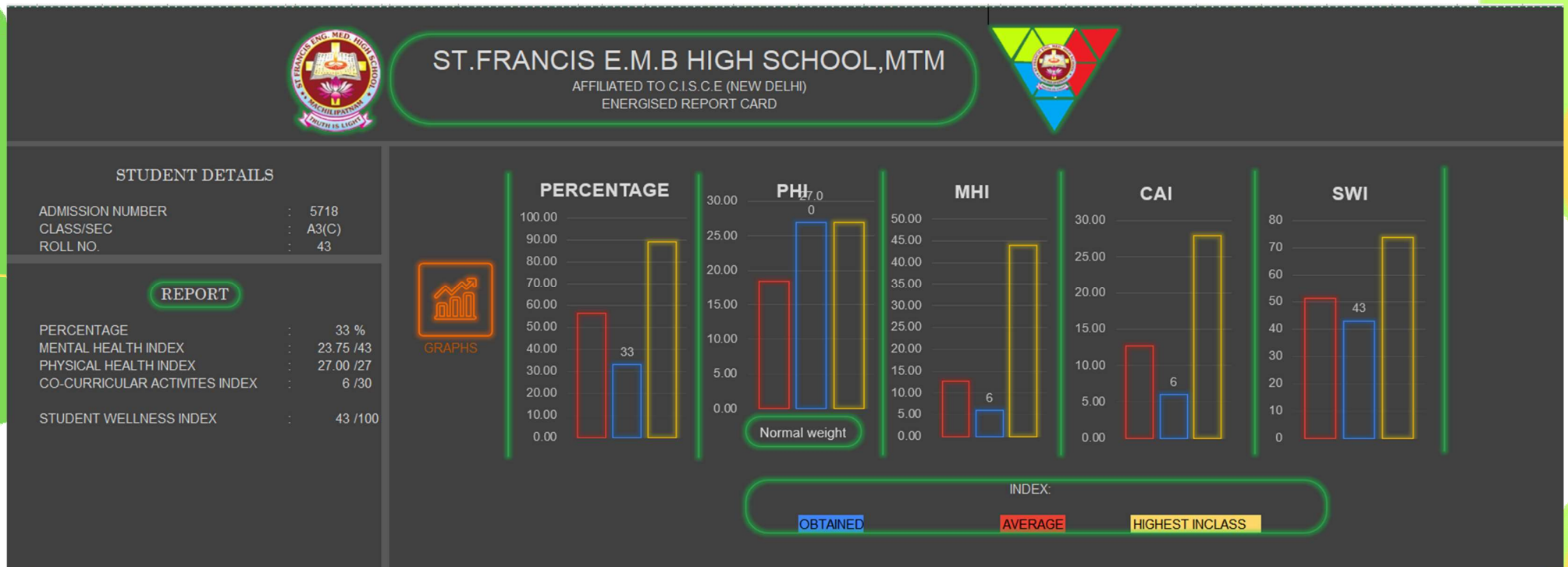
weightage



■ Marks obtained ■ PHI ■ MHI ■ CAI

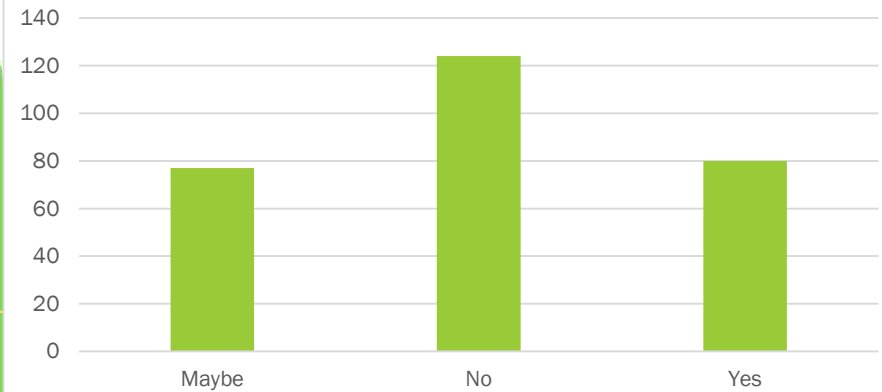
10 Results

With data collected we have developed energized progress report which is as follows



The pie graph shows that more than 50% of students surveyed aren't comfortable in discussing about their mental health with parents and teachers while over 30% students may discuss but not with certainty

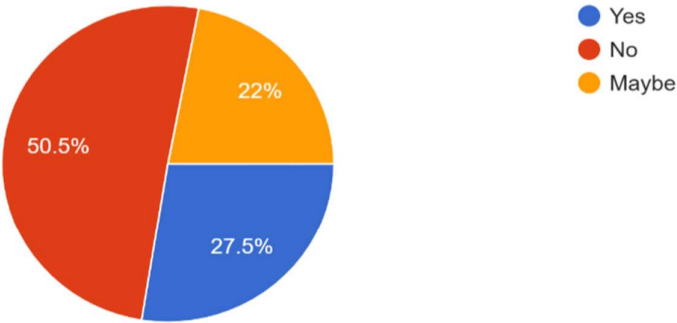
Students who feel comfortable to discuss with parents or teachers



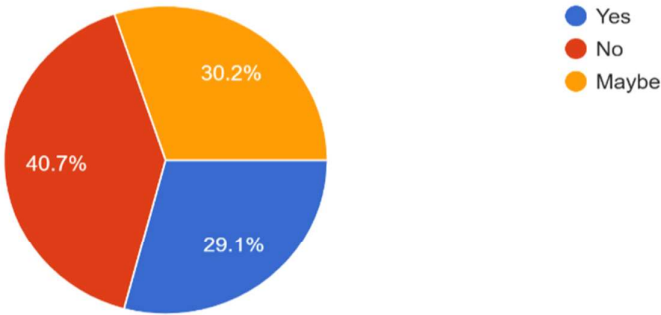
Students who feel comfortable to discuss with parents or teachers	No. of students
Maybe	77
No	124
Yes	80

TABLE 8

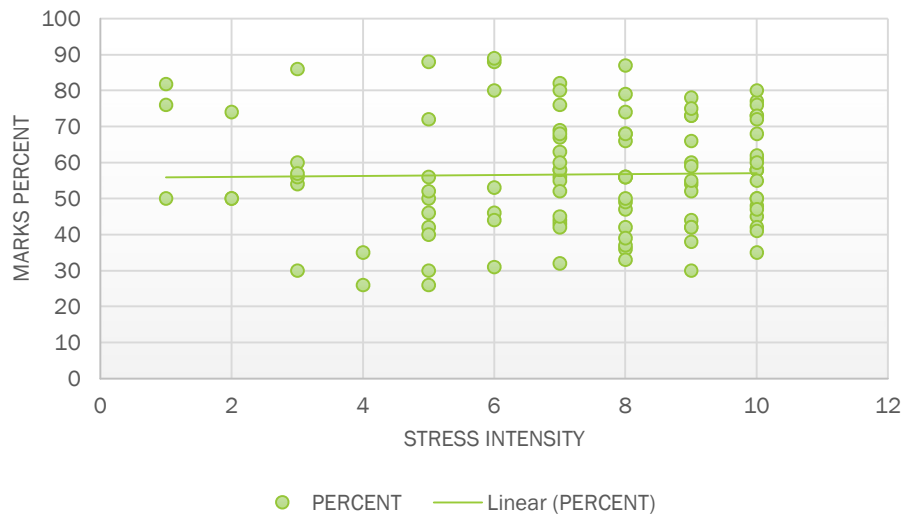
Are you comfortable discussing about Stress with your Parents or teachers
109 responses



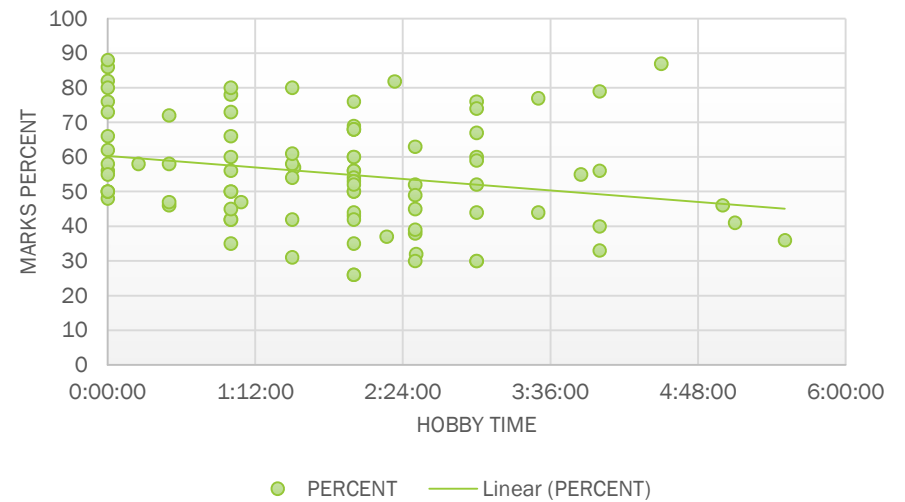
Are you comfortable discussing about Stress with your Parents or teachers
199 responses



PERCENTAGE vs. STRESS INTENSITY



PERCENTAGE vs. HOBBY TIME



PERCENTAGE vs. STRESS INTENSITY

Overview of graph looks very clumsy but if we observe it carefully, we can note that students with the percentage of 80-50% experience more stress than other students we can also take a note that not even one student in range of 70-50% has a stress index of above 6

PERCENTAGE vs. HOBBY TIME

By the trend line we can clearly see that as percentage decreases the hobby time increases so as a child spends more time for a hobby his academic performance decrease so the area with most density of dots is taken as optimum time for hobby is 1hr to 2hrs is derived

11 Discussion of results & Future scope

The results obtained above are discussed with teachers and parents to get a better understanding of role of children school eco system by conducting a meeting among staff and school management.

Due to major constraint of time, we have limited the scope of project. In future this project can be slightly modified and can be done with even diverse sample size including children of 8th and 7th classes and apart from surveying students, a different survey form can also be given to parents and teachers to gather their opinion on this regard. This also can be done once in 6 months to see the development in child within a span of time

12 LIMITATIONS

- Child's fear
- Time is major constraint
- Child's understanding of questions

13 Conclusion

Based on the research findings, in general, students experience stress in the moderate category. This is an ideal condition to encourage students to learn because students get adequate challenges to motivate and show optimal performance in learning. Moreover, there is a need to highlight that they should enjoy the process and optimize the quality of their learning.

Teachers are also expected to be able to support students to increase this capacity so that they do not perceive academic stress as something terrifying that must be avoided but as something to be dealt with using adequate capacity. However, there are some respondents admitting that they often or always experience academic stress. This means that the academic stress faced has the potential to interfere with wellbeing because it is excessive and exceeds the individual's capacity to face it. This condition needs the attention of various parties, the government, students, teachers, and parents. When the number of sources of stress exceeds the individual's capacity, it will cause distress that has a negative impact. The government, especially the Ministry of Education and Culture should continuously improve the quality of education that includes the teacher education system, teacher development, and training, learning process, and materials, as well as an evaluation system to prevent students' academic stress.

14.Acknowledgement

I commence with name of the Almighty "God" benevolence and beneficence who enabled me to complete this project.

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Yours Faithfully,
Yaswanth. Uppalapu
Group leader

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THANK YOU

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